Content:
Elastic Properties of Materials: Types of Stress and Strain, Stress Strain Curve, Stress hardening and softening, Elastic moduli, Relation between Elastic constants, Poisson's ratio and its limiting value, Bending of beams: neutral surface and neutral axis, expression for bending moment of a beam: Single cantilever (derivation). Numerical problems. Torsion of a Cylinder: Expression for couple per unit twist of a solid cylinder, torsion pendulum: expression for time period and rigidity modulus, Numerical problems.

Introduction:
Materials are classified as Rigid, Elastic and Plastic based on the behavior of the material on the application of external force. The applied force is called deforming force.

A rigid body is a body which does not undergo any deformation when external forces act on it. When forces are applied on a rigid body the distance between any two particles of the body will remain unchanged however large the force may be. In actual practice no body is perfectly rigid. For practical purposes solid bodies are taken as rigid bodies.

When a body is acted upon by a suitable force, it undergoes a change in form, this change in form is called **Deformation**. The change could be either in shape or size or even both. If the body recovers its original state on the removal of deforming force, then the body is called an **Elastic body**. eg. Quartz

"Elasticity is the property of the material of a body by virtue of which it regains its original shape and size after the deforming force is removed".

If the body does not show any tendency of returning back to its original or initial state and stays in the changed form after the withdrawal of external force, then it is said to be in Plastic state. eg Clay.

"Plasticity is the property of the material of a body by virtue of which it fails to regain its original shape and size after the deforming force is removed".

What happens to the body when these forces are removed?

The external forces acting on the body compel the molecules to change their position. Due to these relative molecular displacements, internal forces are developed within the body that tends to oppose the external deforming force. When the deforming forces are removed, the internal forces will tend to regain the shape and size of the body.

How does one account for this?

Under the action of the external force, the body changes its form because; the molecules inside it are displaced from their previous positions. While they are displaced, the molecules develop a tendency to come back to their original positions, because of intermolecular binding forces. The aggregate of the restoration tendency exhibited by all the molecules of the body, manifests as a balancing force or restoring force counteracting the external force.

Important terms to be understood

Deforming Force: The applied external force, which causes, deformation is called deforming force.

Restoring force: The reactional force which is developed in a deformed body is called restoring force.

LOAD: The external force acting on a body is the load and its effect is to change the form or the dimensions of the body.

STRESS: It is the ratio of restoring force to the area of its application. Unit of stress is Nm^{-2}.

Normal stress: Restoring force per unit area perpendicular to the surface is called normal stress.

Tangential stress: Restoring force parallel to the surface per unit area is called tangential stress.

STRAIN:

The deformation produced by the external force accompanying a change in dimensions or form of the body is called strain. It is the **ratio of change in dimensions of the body to its original dimensions.**

The way in which the change in dimensions is produced depends upon the form of the body and how the force is applied.

Deformation is of three types, resulting in three types of strains, defined as follows:

i) **Linear strain or Tensile strain**: If the shape of the body could be approximated to the form of a long wire and if a force is applied at one end along its length keeping the other end fixed, the wire undergoes a change in length.

If x is the change in length produced for an original length L then,

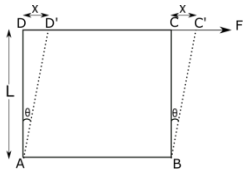
$$\textbf{Linear strain} = \frac{change\ in\ length}{original\ length} = \frac{x}{L}$$

ii) **Volume strain or Bulk strain:** If a uniform force is applied all over the surface of a body, the body undergoes a change in its volume (however the shape is retained in case of solid bodies). If v is the change in volume to an original volume V of the body, then,

$$\textbf{Volume strain} = \frac{\text{change in volume}}{\text{original volume}} = \frac{v}{V}$$

iii) **Shear strain:** If a force is applied tangentially to a free surface of a body with the other surface being fixed, its layers slide one over the other; the body experiences a turning effect and changes its shape. This is called **shearing** and the angle through which the turning takes place is called **shearing angle (θ).**

$$\text{Shearing strain } (\theta) = \frac{x}{L}$$

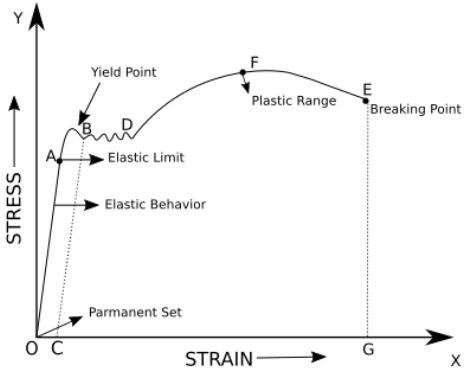


HOOKE'S LAW: This is a relation between stress and strain. It states that **"stress is directly proportional to strain" (provided strain is small), i.e., stress α strain,**

$$\textbf{Or, } \frac{stress}{strain} = \text{a constant (E)}$$

The constant of proportionality E is called the modulus of elasticity or coefficient of elasticity. (This is the measure of elasticity of the material of the body)

STRESS - STRAIN RELATIONSHIP IN A WIRE



A graph showing the variation of stress with strain is called stress-strain diagram.

When the stress is continually increased in the case of a solid, a point is reached at which the strain increases rapidly. The stress at which the linear relationship between stress and strain ceases to hold good is referred to as the *elastic limit of the material*. Thus, if the material happens to be in the form of a bar or a wire under stretch, it will recover its original length on the removal of the stress so long as the stress is below the elastic limit, but if this limit is exceeded, it will fail to do so and will acquire what is called as '*permanent set*'.

The relationship between stress and strain is studied by plotting a graph for various values of stress and the accompanying strain.

The straight and sloping part OA of the curve shows that the strain produced is directly proportional to the stress applied or the Hooke's law is obeyed perfectly up to A and that, therefore, on the removal of the stress, it will recover its original condition of zero strain, represented by O.

As soon as the elastic limit is crossed, the strain increases more rapidly than the stress, and the graph curves along AB, the extension of the wire now being partly elastic and partly plastic. Hence, on being unloaded here, say, at the point B, it does not come back to its original condition along AO, but takes the dotted path BC, so that there remains a residual strain OC in it, which is *permanent set acquired by the wire*. **The strain left behind in the material even after the removal of the applied load is called residual strain.**

Beyond the point B, for practically little or no increase in stress(or the load applied) there is a large increase in strain (i.e., in the extension produced) up to D, so that the portion BD of the graph is an irregular wavy line, the stress corresponding to D being less than that corresponding to B. This point B where the large increase in strain commences is called the **yield point**, the stress corresponding to it being known as the **yielding stress.** The yielding ceases at D and further extension, which now becomes plastic,(being mainly due to shear stress rather than simple tension) can only be produced gradually increasing the load so that the portion DF of the graph is obtained, the cross-section of the wire decreasing uniformly with extension (or strain) upto F and hence volume remaining constant. *The maximum load (or force) to which the wire is subjected divided by its original cross sectional area is called the* ***ultimate strength*** *or the* ***tensile strength*** *of the wire and is also termed as* ***breaking stress****.*

The extension of the wire goes on increasing beyond F without any addition to the load, even if the load is reduced a little, and wire behaves as though it were literally 'flowing down'. This is because of a faster rate of decrease of its cross-sectional area at some section of its length where local constriction, called a "neck" begins to develop,

with the result that, even if the load is not increased, the load per unit area or the stress becomes considerably greater there, bringing about corresponding increase in strain or extension in the wire. The load is then decreased, i.e., the stress reduced at this stage and the wire finally snaps or breaks at E, which thus represents the breaking point for it.

Elastic Fatigue:

We recall that, if the maximum stress in the specimen does not exceed the elastic limit of the material, the specimen returns to its initial condition when the load is removed. Such a conclusion is correct for loadings repeated a few dozen or even a few hundred times. However, it is not true when loading is repeated thousands or millions of times. In such a case rupture will occur at a stress much lower than the static breaking strength, this phenomenon is known as Fatigue. A fatigue failure is of a brittle nature even for materials that are normally ductile.

A body subjected to repeated strains beyond its elastic limit, has its elastic properties greatly impaired, and may break under a stress less than its normal breaking stress even within its elastic limit. This is called elastic fatigue.

Fatigue must be considered in the design of all structural and machine components that are subjected to repeated or to fluctuating loads. The number of loading cycles that may be expected during the useful life of a component varies greatly. For example, a beam supporting an industrial crane may be loaded as many as two million times in 25 years (about 300 loadings per working day) an automobile crankshaft will be loaded about half a billion times if the automobile

is driven 320,000 km, and an individual turbine blade may be loaded several hundred billion times during its lifetime.

Strain Hardening:

When a metal is stressed beyond its elastic limit, it enters the plastic region (in which the residual strain remains upon unloading). When the load is further increased, the material hardens and becomes stronger, i.e it's more difficult to deform the material as the strain increases and hence it's called "strain hardening". This tends to increase the strength of the material and decreases its ductility. It is region between yield point and ultimate tensile strength.

Strain softening:

Strain softening is defined as the region in which the stress in the material is decreasing with an increase in strain. It is observed after the yield point. This is due to brittleness and heterogeneity of the material.

Corresponding to the three types of strain, we have three types of elasticity:

a) **Linear Elasticity or Young's modulus (Y):**

When the deforming force is applied to the body, along a particular direction, the change per unit length in that direction is called longitudinal or linear or elongation strain, and the force applied per unit area of cross section is called longitudinal or linear stress. **The ratio of longitudinal stress to linear strain within elastic limit is called the Young's modulus and is denoted by Y.**

If F is the force applied normally, to a cross-sectional area 'a', then the stress is F/a. If L is original length and x is change in length due to the applied force, the strain is given by x/L, so that,

$$\mathbf{Y} = \frac{\text{Normal stress}}{\text{Longitudinal strain}} = \frac{F/a}{x/L} = \frac{FL}{ax} \quad \mathbf{N/m^2}$$

b) **Bulk modulus(K):**

When the deforming force is applied uniformly along the normal to the surface of a body, it produces a volume strain (without changing its shape in case of solid bodies). The applied force per unit area gives the normal stress or pressure. **The ratio of the normal stress to the volume strain within elastic limit is called the Bulk modulus and is denoted by K.**

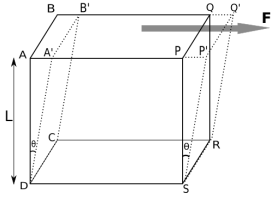
If F is the force applied uniformly and normally on a surface area 'a', the stress or pressure is *F/a or P* and if *v* is the change in volume produced in an original volume *V*, the strain is given by *v/V* and therefore

$$\mathbf{K} = \frac{\text{Normal stress}}{\text{Volume strain}} = \frac{F/a}{v/V} = \frac{FV}{av} = \frac{PV}{v} \quad \mathbf{N/m^2}$$

NOTE: Bulk modulus is referred to as incompressibility, and hence its reciprocal is called compressibility (strain per unit stress).

c) **Modulus of Rigidity (η):**

In this case, while there is a change in the shape of the body, there is no change in its volume. It takes place by the movement of layers of the body, one over the other.



Consider a rectangular solid cube whose lower face DCRS is fixed, and to the upper face a tangential force F is applied in the direction as shown. Under the action of this force, the layers of the cube which are parallel to the applied force slide one over the other such that point A shifts to A^1, B to B^1, P to P^1 and Q to Q^1. The body has turned through an angle θ. This angle θ is called the angle of shear or **shearing**

strain. Tangential stress is equal to the force F divided by area 'a' of the face APQB.

Hence **tangential stress** = $\frac{F}{a}$ Shearing strain (θ) = PP^1 / PS = $\frac{x}{L}$

The rigidity modulus is defined as the ratio of the tangential stress to the shearing strain within elastic limit is called the rigidity modulus and is denoted by η.

Rigidity modulus η = $\frac{\text{tangential stress}}{\text{shearing strain.}} = \frac{F/a}{\theta} = \frac{F/a}{x/L} = \frac{FL}{ax}$ **N/m²**

Poisson's Ratio (σ):

In case of any deformation taking place along the length of a body like a wire, due to a deforming force, there is always some change in the thickness of the body. This change which occurs in a direction perpendicular to the direction along which the deforming force is acting is called **lateral change**.

Within elastic limits of a body, the ratio of lateral strain to the longitudinal strain is a constant and is called Poisson's ratio.

If a deforming force acting on a wire of length *L* produces a change in length '*x*' accompanied by a change in diameter '*d*' in it which has a original diameter of *D*, then lateral strain $\beta = \frac{d}{D}$ and Longitudinal strain $\alpha = \frac{x}{L}$, ∴ **Poisson's ratio, σ** = $\frac{\beta}{\alpha} = \frac{Ld}{xD}$

There are no units for Poisson's ratio. It is a dimensionless quantity.

Relation between the three moduli of elasticity:

When a body undergoes an elastic deformation, it is studied under any of the three elastic modulii depending upon the type of deformation. However, these modulii are related to each other. Now, their relation can be understood by knowing how one type of deformation could be equated to combination of other types of deformation.

1. Relation between Young's modulus (Y) and elongation strain (α)

Consider a cube of unit side subjected to unit tension along one side. Let α be the elongation per unit length per unit tension along the direction of the force. Therefore, $\text{Stress} = \frac{\text{Force}}{\text{Area}} = 1$

Similarly, linear strain $= \frac{\text{change in length}}{\text{original length}} = \frac{\alpha}{1} = \alpha$

$$\therefore \quad \mathbf{Y} = \frac{\text{Linear stress}}{\text{Longitudinal strain}} = \frac{1}{\alpha}$$

2. Relation between Young's modulus (Y),Bulk modulus (K) and Poisson's ratio (σ)

Let α be the elongation per unit length per unit stress along the direction of the forces and β be the contraction per unit length per unit stress in a direction perpendicular to the respective forces.

$$K = \frac{1}{3\alpha(1-2\beta/\alpha)} \qquad \sigma = \frac{\beta}{\alpha} \qquad \frac{1}{\alpha})$$

$$K = \frac{(1/\alpha)}{3(1-2\beta/\alpha)}$$

$$K = \frac{Y}{3(1-2\sigma)} \qquad (\qquad \text{and } Y =$$

3. Relation between Rigidity modulus (η), Young's modulus (Y) and Poisson's ratio (σ)

$$\eta = \frac{1}{2(\alpha+\beta)} = \frac{1}{2\alpha(1+\beta/\alpha)} = \frac{1/\alpha}{2(1+\sigma)}, \quad (\because \sigma = \beta/\alpha)$$

$$\eta = \frac{Y}{2(1+\sigma)} \quad or \quad Y = 2\eta(1+\sigma)$$

4. Relation between the elastic constants Young's modulus (Y), Bulk modulus (K) and Rigidity modulus (η)

We know that $\eta = \frac{Y}{2(1+\sigma)}$ and K $= \frac{Y}{3(1-2\sigma)}$

Rearranging which we get, $\frac{Y}{\eta} = 2+2\sigma$ and $\frac{Y}{3K}$ =1-2σ

Adding the above we get, $\frac{Y}{\eta} + \frac{Y}{3K} = 3$ or $\frac{Y(3K+\eta)}{3\eta K} = 3$

$\therefore Y = \left[\frac{9\eta K}{3K+\eta}\right]$ Relation connecting the three elastic constants.

5. Relation between Bulk modulus (K),Rigidity modulus (η) and Poisson's ratio (σ)

We have the relations Y = 2η (1 + σ), and Y = 3K (1-2σ)

Equating the above equations we get,

$$2\eta + 2\eta\sigma = 3K - 6K\sigma,$$
$$\therefore 2\eta\sigma + 6K\sigma = 3K - 2\eta,$$
$$\sigma(2\eta + 6K) = 3K - 2\eta,$$
$$\sigma = \frac{3K - 2\eta}{2\eta + 6K}$$

Limiting value of Poisson's ratio:

We know that, $Y=2\eta(1+\sigma)$ and $Y=3K(1-2\sigma)$

$\therefore$ $2\eta(1+\sigma) = 3K(1-2\sigma)$ Where K and η are positive quantities.

(i) If σ be *positive quantity*, then the right hand side and left hand side expression must be positive, and for this to be so, $2\sigma<1$, or $\sigma<0.5$

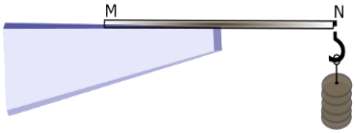
(ii) If σ be *negative quantity*, the left hand expression, and hence also the right hand expression, must be positive, and this is possible only when σ be not less than -1.

Thus the limiting values of σ are -1 and 0.5. A negative value of σ means that a body on being extended should also expand laterally and one can hardly expect this to happen (we know of no substance so far). Similarly, a value of $\sigma=0.5$ would mean that the substance is *perfectly incompressible*, and we do not know of any such substance either.

In actual practice, the value of σ is found to lie between 0.2 and 0.4.

Bending of beams:

A homogenous body of uniform cross section whose length is large compared to its other dimensions is called a beam.

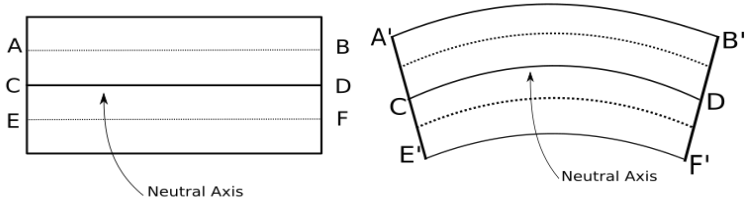


Neutral surface and neutral axis:

Consider a uniform beam MN whose one end is fixed at M. The beam can be thought of consisting of a number of parallel layers and each layer in turn can be thought of as made up of a number of infinitesimally thin straight parallel longitudinal filaments or fibers arranged one closely next to the other in the plane of the layer. If the cross section of the beam along its length and perpendicular to these layers is considered, the filaments of different layers appear like straight lines piled one above the other. For a given layer, all its constituent filaments are assumed to undergo identical changes when that layer is strained.

If a load is attached to the free end of the beam, the beam bends. The successive layers along with constituent filaments are strained. A filament like AB of an upper layer will be elongated to A^1B^1 and the one like EF of a lower layer will be contracted to E^1F^1. But there will always be a particular layer whose filaments do not change their length as shown for CD. Such a layer is called **neutral surface** and the line along which a filament of it is situated is called **neutral axis.**

The filaments of a neutral surface could be taken as line along which the surface is intercepted by a cross section of the beam considered in the plane of bending as shown.



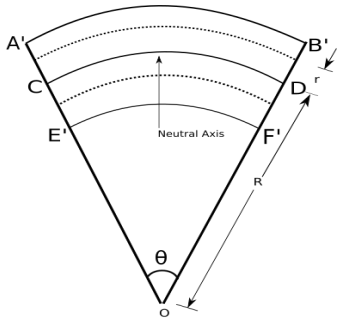
Neutral Surface: It is that layer of a uniform beam which does not undergo any change in its dimensions, when the beam is subjected to bending within its elastic limit.

Neutral axis: It is a longitudinal line along which neutral surface is intercepted by any longitudinal plane considered in the plane of bending.

When a uniform beam is bent, all its layers which are above the neutral surface undergo elongation whereas those below the neutral surface are subjected to compression. As a result, the forces of reaction are called into play in the body of the beam which develops an inward pull towards the fixed end for all the layers above the neutral surface and an outward push directed away from the fixed end for all layer below the neutral surface. These two groups of forces result in a restoring couple which balances the applied couple acting on the beam. The moment of restoring forces about the neutral axis is called **bending moment**. When the beam is in equilibrium, the bending moment and the restoring moments are equal.

Bending moment of a beam:

Consider a uniform beam whose one end is fixed. If now a load is attached to the beam, the beam bends. The successive layers are now strained. A layer like AB which is above the neutral surface will be elongated to A^1B^1 and the one like EF below neutral surface will be contracted to E^1F^1. CD is neutral surface which does not change its length.



The layers of the bent beam can be imagined to form part of concentric circles of varying radii. Let R be the radius of the circle to which the neutral surface forms a part.

$$\therefore CD = R\theta$$

Where, θ is the common angle subtended by the layers at common center O of the circles. The layer AB has been elongated to A^1B^1.

$$\therefore \text{ Change in length} = A^1B^1 - AB$$

$$\text{But } AB = CD = R\theta$$

If the successive layers are separated by a distance r then,

$A^1B^1 = (R+r)\theta$

$\therefore$ Change in length= $(R+r)\theta - R\theta = r\theta$

But original length = $AB = R\theta$

$$\therefore \text{Linear strain} = \frac{r\theta}{R\theta} = \frac{r}{R}$$

Young's Modulus Y= Longitudinal stress / Linear strain

$$\text{Longitudinal stress} = Y \times \text{Linear strain} = Y \times \frac{r}{R}$$

$$\text{But stress} = \frac{F}{a}$$

Where, 'F' is the force acting on the beam and 'a' is the area of the layer AB.

$$\frac{F}{a} = \frac{Yr}{R}$$

$$F = \frac{Yar}{R}$$

Moment of this force about the neutral axis = F x its distance from neutral axis.

$$= F \times r = Yar^2 / R$$

Moment of forces acting on the entire beam

$$\text{Bending moment} = \Sigma \frac{Yar^2}{R} = \frac{Y}{R}\Sigma ar^2$$

The moment of inertia of a body about a given axis is given by Σmr^2, whereas Σar^2 is called the geometric moment of Inertia Ig.

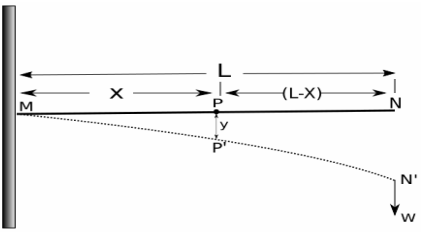
$Ig = \Sigma ar^2 = Ak^2$, where 'A' is the area of cross section of the beam and 'k' is the radius of gyration about the neutral axis.

Bending moment: It is defined as the algebraic sum of moments of restoring forces about neutral axis.

$$\text{Bending moment} = \frac{Y}{R} Ig$$

The quantity YI_g is known as flexural rigidity of the beam and is defined as the bending moment of the beam which produces unit radius of curvature.

Single Cantilever:



If one end of beam is fixed to a rigid support and its other end loaded, then the arrangement is called **single cantilever or cantilever.**

Consider a uniform beam of length 'L' fixed at M. Let a load 'W' act on the beam at N.

Consider a point on the free beam at a distance 'X' from the fixed end which will be at a distance (L-X) from N. Let P' be its position after the beam is bent.

$\therefore$ Bending moment = Force x Perpendicular distance = W (L-X)

But bending moment of a beam is given by $\frac{Y}{R} Ig$

Where R is the radius of curvature of the bent beam, Y is the young's modulus Ig is the geometric moment of inertia

$$\frac{Y}{R} Ig = W(L-x) \text{ ----------(1)}$$

(1/R is the curvature of the neutral axis)

$$\frac{1}{R} = \frac{W(L-x)}{YIg} \text{ ----------(2)}$$

But if y is the depression of the point P, then it can be shown that

$$\frac{1}{R} = \frac{d^2y}{dx^2} \text{ ----------(3)}$$

Comparing equations (2) and (3)

$$\frac{d^2y}{dx^2} = \frac{W(L-x)}{Y\,Ig}$$

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{w(L-x)}{Y\,I_g}$$

Integrating both sides

$$\frac{dy}{dx} = \frac{w}{YI_g}\left[Lx - \frac{x^2}{2}\right] + C_1$$

C_1 is constant of integration

----------(4)

But dy/dx is the slope of the tangent drawn to the bent beam at a distance x from the fixed end. When x=0, it refers to the tangent drawn at M, where it is horizontal. Hence dy/dx=0 at x=0. Introducing this condition in equation (4) we get C_1=0

Equation 4 becomes

$$\frac{dy}{dx} = \frac{W}{YIg}\left[Lx - \frac{x^2}{2}\right]$$

$$dy = \frac{W}{YIg}\left[Lx - \frac{x^2}{2}\right]dx$$

Integrating both sides we get

$$y = \frac{W}{Y\,Ig}\left[\frac{Lx^2}{2} - \frac{x^3}{6}\right] + C_2 \quad \text{---------- (5)}$$

Where, C_2 is constant of integration, y is the depression produced at known distance from the fixed end. Therefore, when x=0, it refers to the depression at M, where there is obviously no depression. Hence y=0 at x=0. Introducing this condition in equation (5) we get

$$y = \frac{W}{YIg}\left[\frac{Lx^2}{2} - \frac{x^3}{6}\right]$$

At the loaded end, y = δ and x = L , Therefore $\delta = \frac{W}{YIg}\left[\frac{L^3}{2} - \frac{L^3}{6}\right]$

Depression produced at loaded end is

$$\delta = \frac{WL^3}{3YIg}$$

Therefore, the young's modulus of the material of the cantilever is

$$Y = \frac{W\,L^3}{3\,\delta\,Ig} \quad \text{----------(6)}$$

Case (a):

If the beam is having rectangular cross-section, with breadth b and thickness d then, $Ig = \frac{bd^3}{12}$ ----------(7)

Substituting equation (7) in equation (6) we get

$$Y=\frac{WL^3}{3\delta}\times\frac{12}{bd^3}$$

$$Y=\frac{4\,W\,L^3}{\delta\,b\,d^3}$$

Case(b):

If the beam is having a circular cross section of radius r then,

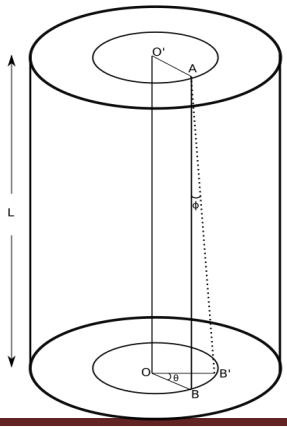
$$Ig=\frac{\Pi r^4}{4} \text{ ----------(8)}$$

Substituting equation (8) in equation (7) we get

$$Y=\frac{4WL^3}{\Pi y\ r_0^{\ 4}}$$

Torsion of a cylinder/Torsion of a shaft

A long body which is twisted around its length as an axis is said to be under torsion. The twisting is brought into effect by fixing one end of the body to a rigid support and applying a suitable couple at the other end. The elasticity of a solid, long uniform cylindrical body under torsion can be studied, by imagining it to be consisting of large number of coaxial, hollow cylinders one within the other. The applied twisting couple is calculated in terms of the rigidity modulus of the body.



Expression for twisting couple or Torsion of a cylindrical rod:

Consider a long cylindrical rod of length 'L' and radius 'R'. Its upper end is clamped and the rod is twisted through an angle θ, by applying a couple to its lower end in a plane perpendicular to its length.